

What is HTML?

HTML stands for Hyper Text Markup Language.

It is used for creating a web pages, and web applications.

It describes the structure of web pages.

It consist of series of elements.

HTML can be invented by Tim Berners Lee in 1992.

Example :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<title> Page Title </title>
```

```
</head>
```

```
<body>
```

```
<h1> My first heading </h1>
```

```
<p> My first paragraph </p>
```

```
</body>
```

```
</html>
```

Explanation :-

<!doctype> :-

It defines that this document is an HTML document.

<html> :-

It is the root element of an HTML pages.

<head> :-

It contains meta information about HTML page.

<title> :-

It specifies a title for the HTML page [which is shown in the browser's title bar or in page tag].

`<body> :-`

It defines the documents body and is container for all the visible contents, such as heading, paragraph, images, hyperlinks, tables, lists, etc.

`<h1> :-`

It defines large heading.

`<p> :-`

It defines paragraph.

- What is HTML element?

HTML elements defines by a starting tag, some content and ending tag.

Syntax :-

`<tagname> ----- content goes here ----- </tagname>`

HTML Attributes

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Attributes provides additional information about the HTML elements.

It always specified in the start tag.

It usually come in name or value pair like name = "value"

- href attribute :-

<a> tag defines a hyperlink. The href attribute specifies the URL of the page, link goes to.

- Example :-

```
<!doctype>
<html>
<body>
<h2> The href attribute </h2>
<a href = "https://www.w3schools.com" > Visit w3
schools </a>
</body>
</html>
```

- src attribute :-

The tag is used to embed an image in an HTML page. The src attribute specifies the path to the image to be displayed.

- Example :-

```
<!doctype>
<html>
<body>
<h2> The src attribute </h2>
<img src = "img.jpg" width = "500" height = "500">
</body>
</html>
```

- Two ways to specify the URL in the src attribute :-

i. Absolute URL :-

Link to an external image that is hosted on the another website.

ii. Relative URL :-

Links to an image that is hosted within the website.

- Width and height attributes :-

The tag should also contain the width and height attributes, which specifies the width and height of image [in pixels].

Syntax :-

<element attribute-name = "value" > content </element>

- Example :-

```
<!doctype >
```

```
<html >
```

```
<body >
```

```
<h1 > This is style attribute </h1 >
```

```
<h1 title = " This is heading tag" > Move the cursor over the  
heading and paragraph </h1 >
```

```
<p style = "height : 50px ; color : blue" > It will add style  
property in element </p >
```

```
<p style = "color : red" > It will change the color of content </p >
```

```
<img src = "flower.jpg" alt = "flower" height = "200" width =  
"200" >
```

```
</body >
```

```
<img src = "https://static.javatpoint.com/htmlpages/  
images/whitepeacock.jpg" height = "200" width = "200" >
```


<b1>

 This is link

</body>

</html>

- Example :-

<!doctype>

<html>

<body>

<h1>Heading </h1>

<hr>

<h1 style = "font-size : 60px ;" > Heading1 </h1>

</body>

</html>

HTML style

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HTML style is used to change or add the style on existing HTML elements.

The default style for every HTML element e.g. background color is white, text color is black.

The style attribute can be used with any HTML tag.

Syntax:-

style = "property: value"

- Example :-

```
<!doctype>
```

```
<html>
```

```
<body style = "background-color: purple;">
```

```
<h2> Welcome </h2>
```

```
<h2 style = "color: red;"> This is HTML </h2>
```

```
<h2 style = "color: blue;"> Hello </h2>
```

```
<p style = "font-size: 50px;"> Hi </p>
```

```
<p style = "font-size: 160%;"> This is a paragraph </p>
```

```
<h1 style = "background-color: powderblue;"> This is a  
heading </h1>
```

```
<h1 style = "background-color: tomato; color: white;">
```

```
  This is disha computer institute </h1>
```

```
<h2 style = "font-family: arial;"> Welcome to India </h2>
```

```
<h3 style = "text-align: center;"> centered heading </h3>
```

```
<h3 style = "text-align: center;"> centered paragraph </h3>
```

```
</body>
```

```
</html>
```

HTML Formatting

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It is a process of formatting text for better look. HTML provides us ability to format text without using CSS.

In HTML the formatting tags are divided into two categories.

i. Physical tag :-

These tags are used provide the visual appearance to the text.

ii. Logical tag :-

These tags are used to add some logical or semantic value to the text.

- There are some formatting tags.

** :-**

It is used to bold the text written between it.

** :-**

It tells the browser that the text is important.

<i> :-

It is used to make text italic.

<mark> :-

It is used to highlight the text.

<u> :-

It is used to underline the text written between it.

<tt> :-

It is used to appear the text in teletype (monospace) format.

<strike>:-

It is used to draw a strike through on a section of a text.

<sup>:-

It displays the content slightly above the normal line.

<sub>:-

It displays the content slightly below the normal line.

:-

It is used to display the deleted content.

<ins>:-

It displays the content which is added.

<big>:-

It is used to increase the font size by one conventional unit.

<small>:-

It is used to decrease the font size by one unit from base font size.

- **Example :-**

<!doctype>

<body>

<p>Write paragraph in bold text </p>

**<p>This is an important content **
and this is normal content **</p>**

<p><i>Write paragraph in italic text </i></p>


```
<p><em> This is an important content </em>, which
displayed in italic font </p>
</body>
</html>
```

- Example :-

```
<!doctype>
<html>
<body>
<h2> I am from <mark> India </mark> </h2>
<p><u> Write a paragraph in underline text </u> </p>
<p><strike> Write a paragraph with strike </strike> </p>
<p>Hello <tt> Write paragraph in monospace font </tt> </p>
<p>Hello <sup> Write text in superscript </sup> </p>
<p>Hello <sub> Write text in subscript </sub> </p>
</body>
</html>
```

Example :-

```
<!doctype>
<html>
<body>
<p>Hello <del> delete your paragraph </del> </p>
<p><del> delete content </del> <ins> and write a new
content </ins> </p>
<p>Hello <big> write a paragraph in larger font </big> </p>
<p>Hello <small> write a paragraph in the smaller font
</small> </p>
</body>
</html>
```

HTML Quotation

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- blockquote tag :-

It is used to define a large quoted section. If you have a large quotation then put the entire text within the blockquote tag.

- Example :-

```
<!doctype>
<html>
<body>
<p> Learn about HTML : </p>
<blockquote> HTML stands for Hyper Text Markup
language. It is used to create web pages and web
applications.
It is widely used language on the web. </blockquote>
</body>
</html>
```

- <q> for short quotation :-

It is used to put small quotation.

- Example :-

```
<!doctype>
<html>
<body>
<p> HTML stands for : </p>
<q> Hyper Text Markup language </p>
</body>
</html>
```

- <abbr> for Abbreviations :-

It can give useful information to browser, translation

System and search engines.

- Example :-

```
<!doctype>
<html>
<body>
<p> The <abbr title = "World Health organization">WHO
    </abbr> was founded in 1948 </p>
</body>
</html>
```

- <address> for contact information :-

It defines the contact information for the author/owner of a document or an article.

contact information can be an email address, URL, office address, phone number, social media handle etc.

The text in <address> element usually in italic, and the browser will always add a line break before and after the <address> element.

- Example :-

```
<!doctype>
<html>
<body>
<p> Address element defines contact information </p>
<address>
Written by Disha computer <br>
Visit us at: <br>
Example.com <br>
Box 564, Ahmednagar <br>
Maharashtra
```

</address>

</body>

</html>

- <cite> for work title :-

It defines the title of creative work [e.g. book, poem, song, movie, painting, etc].

- Example :-

<!doctype>

<html>

<body>

<img src = "flower.jpg" width = "220" height = "277"
alt = "flower">

<p><cite> created by </cite> xyz - painted in 2010 </p>

</body>

</html>

- <bdo> for bi-directional override :-

It is used to override the current text direction.

- Example :-

<!doctype>

<html>

<body>

<bdo dir = "rtl"> This is will be written from right to
left </bdo>

</body>

</html>

Comments

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Comments are some text or code written in your code to give an explanation about the code, and not visible to the user.

Comments which are used for HTML file are known as HTML comments.

There are two types of comments.

i. single line comments :-

```
<!-- Write commented text here -->
```

ii. multi-line comments :-

```
<!-- Your code is commented
```

```
Write description of code
```

```
It will not display at web page
```

```
-->
```

- RGB and RGBA colors :-

- Shades of colors :- (RGB colors)

```
<!doctype>
```

```
<html>
```

```
<body>
```

```
<h1 style="background-color: rgb(60, 60, 60);"> Hello  
World </h1>
```

```
<h1 style="background-color: rgb(100, 100, 100);"> Hello  
World </h1>
```

```
<h1 style="background-color: rgb(140, 140, 140);"> Hello  
World </h1>
```

```
<h1 style="background-color: rgb(180, 180, 180);"> Hello  
World </h1>
```

```
<h1 style="background-color: rgb(200, 200, 200);"> Hello  
World </h1>
```

```
<h1 style="background-color: rgb(240, 240, 240);"> Hello  
World </h1>
```

```
</body>
```

```
</html>
```

- RGBA colors :-

```
<!doctype>
```

```
<html>
```

```
<body>
```

```
<h1 style="background-color: rgba(255, 99, 71, 0);">  
Hello World </h1>
```

```
<h1 style="background-color: rgba(255, 99, 71, 0.2);">  
Hello World </h1>
```

```
<h1 style="background-color: rgba(255, 99, 71, 0.4);">  
Hello World </h1>
```

```

<h1 style = "background-color: rgba(255, 99, 71, 0.6);">
  Hello World </h1>
<h1 style = "background-color: rgba(255, 99, 71, 0.8);">
  Hello World </h1>
<h1 style = "background-color: rgba(255, 99, 71, 1);">
  Hello World </h1>
</body>
</html>

```

- HEX colors :-

Hexadecimal color is specified with : #RRGGBB
Hexadecimal integer specify the components of the color.

- Example :-

```

<!doctype>
<html>
<body>
<h1 style = "color: #ff0000;"> Welcome </h1>
<h1 style = "color: #0000ff;"> Welcome </h1>
<h1 style = "color: #3cb371;"> Welcome </h1>
<h1 style = "color: #ee82ee;"> Welcome </h1>
<h1 style = "color: #ffa500;"> Welcome </h1>
<h1 style = "color: #6a5acd;"> Welcome </h1>
</body>
</html>

```

- shades of HEX colors :-

```

<!doctype>
<html>
<body>

```

```

<h1 style = "background-color: # 404040;" > Welcome </h1>
<h1 style = "background-color: # 686868;" > Welcome </h1>
<h1 style = "background-color: # a0a0a0;" > Welcome </h1>
<h1 style = "background-color: # bebebe;" > Welcome </h1>
<h1 style = "background-color: # dcdcdc;" > Welcome </h1>
<h1 style = "background-color: # f8f8f8;" > Welcome </h1>
</body>
</html>

```

- HSL and HSLA color :-

HSL stands for Hue, Saturation and Lightness.

- Example :-

```

<!doctype>
<html>
<body>
<h1 style = "color: hsl (0, 100%, 50%);" > Hello </h1>
<h1 style = "color: hsl (240, 100%, 50%);" > Hello </h1>
<h1 style = "color: hsl (147, 50%, 47%);" > Hello </h1>
<h1 style = "color: hsl (300, 76%, 72%);" > Hello </h1>
<h1 style = "color: hsl (39, 100%, 50%);" > Hello </h1>
<h1 style = "color: hsl (248, 53%, 58%);" > Hello </h1>
</body>
</html>

```

- Saturation :-

Saturation can be described as the intensity of color.

100% is pure color no shade of color

50% is 50% gray but you can still see the color.

0% is completely gray. you can no longer to see color.

Example :-

```
<!doctype>
<html>
<body>
<h1 style="background-color: hsl(0, 100%, 50%);"> Hello
World </h1>
<h1 style="background-color: hsl(0, 80%, 50%);"> Hello
World </h1>
<h1 style="background-color: hsl(0, 60%, 50%);"> Hello
World </h1>
<h1 style="background-color: hsl(0, 40%, 50%);"> Hello
World </h1>
<h1 style="background-color: hsl(0, 20%, 50%);"> Hello
World </h1>
<h1 style="background-color: hsl(0, 0%, 50%);"> Hello
World </h1>
</body>
</html>
```

- Lightness :-

It describes how much light you want to give the color, where 0% means no light (black), 50% means 50% light (neither dark nor light), 100% means full lightness (white).

Example :-

```
<!doctype>
<html>
<body>
<h1 style="background-color: hsl(0, 100%, 0%);">
Hello World </h1>
```

```

<h1 style="background-color: hsl(0, 100%, 25%);">
  Hello World </h1>
<h1 style="background-color: hsl(0, 100%, 50%);">
  Hello World </h1>
<h1 style="background-color: hsl(0, 100%, 75%);">
  Hello World </h1>
<h1 style="background-color: hsl(0, 100%, 90%);">
  Hello World </h1>
<h1 style="background-color: hsl(0, 100%, 100%);">
  Hello World </h1>
</body>
</html>

```

- Shades in HSL color :-

```

<!doctype>
<html>
<body>
<h1 style="background-color: hsl(0, 0%, 20%);">
  Welcome </h1>
<h1 style="background-color: hsl(0, 0%, 30%);">
  Welcome </h1>
<h1 style="background-color: hsl(0, 0%, 40%);">
  Welcome </h1>
<h1 style="background-color: hsl(0, 0%, 60%);">
  Welcome </h1>
<h1 style="background-color: hsl(0, 0%, 70%);">
  Welcome </h1>
<h1 style="background-color: hsl(0, 0%, 90%);">
  Welcome </h1>
</body>
</html>

```

- HSLA color values :-

HSLA color values are an extension of HSL color values with an Alpha channel which specifies the opacity for a color.

Example :-

```
<!doctype>
<html>
<body>
<h1 style="background-color: hsla(9, 100%, 64%, 0);">
    hsla color </h1>
<h1 style="background-color: hsla(9, 100%, 64%, 0.2);">
    hsla color </h1>
<h1 style="background-color: hsla(9, 100%, 64%, 0.4);">
    hsla color </h1>
<h1 style="background-color: hsla(9, 100%, 64%, 0.6);">
    hsla color </h1>
<h1 style="background-color: hsla(9, 100%, 64%, 0.8);">
    hsla color </h1>
<h1 style="background-color: hsla(9, 100%, 64%, 0.9);">
    hsla color </h1>
</body>
</html>
```

CSS stands for cascading style sheets.
CSS saves a lot of work. It can control layout of multiple web pages all at once.

- What is CSS :-

It is used to format the layout of a webpage.
With CSS, you can control the color, font, size of text, spacing between elements, how elements are positioned and layout, what background image and background color are to be used, different displays for different devices and size and all.

The CSS can be added to HTML documents in 3 ways:

i. Inline :-

By using the `<style>` attribute inside HTML elements.

ii. Internal :-

By using `<style>` element in the `<head>` section.

iii. External :-

Define all the CSS properties in a separate .css file, and then include file with HTML file using tag in section.
or by using `<link>` element to link an external CSS file.

- Inline CSS :-

It is used to apply a unique style to a single HTML element.
An inline CSS uses the style attribute of an HTML element.
We can use as many as properties as we want and each property should be separated by a semicolon [;].

- Example :-

```
<!doctype>
<html>
<body>
<h1 style = " color : blue ; " > A blue heading </h1>
<p style = " color : red ; " > A red paragraph </p>
</body>
</html>
```

- Example :-

```
<!doctype>
<html>
<head>
<title> Inline CSS </title>
</head>
<body>
<h3 style = " color : red ; font - style : italic ; text - align :
center ; font - size : 50px ; background - color :
gray ; padding - top : 25px ; " >
Learning HTML using Inline CSS </h3>
</body>
</html>
```

- Internal CSS :-

It is used to define a style for a single HTML page.
An internal CSS is defined in the <head> section of an HTML page, within a <style> element.

- Example :-

```
<!doctype>
<html>
<head>
```

```

<style>
body
{
    background-color: violet;
}
h1
{
    color: blue;
}
p
{
    color: red;
}

```

```

</style>
</head>
<body>
<h1> This is a heading </h1>
<p> This is paragraph </p>
</body>
</html>

```

- Example :-

```

<!doctype>
<html>
<head>
<style>
/* Internal CSS using element name */
body
{
    background-color: lavender;
    text-align: center;
}

```

```

h2
{
    font-style: italic;
    font-size: 30px;
    color: #f08080;
}

p
{
    font-size: 20px;
}

/* Internal CSS using class name */
.blue
{
    color: blue;
}

.red
{
    color: red;
}

.green
{
    color: green;
}

</style>
</head>
<body>
<h2> Learning HTML with internal CSS </h2>
<p class="blue"> This is blue color paragraph </p>
<p class="red"> This is red color paragraph </p>
<p class="green"> This is green color paragraph </p>
</body>
</html>

```

- External CSS :-

It is used to define the style for many HTML pages. To use an external style sheet add a link to it in the `<head>` section of each HTML pages.

- External file :-

External file can be written in any text editor. The file must not contain any HTML code and must be saved with .css extension.

File 1 :- saved by .css extension.
styles.css

```
body
{
    background-color: pink;
}
h1
{
    color: blue;
}
p
{
    color: purple;
}
```

File 2 :-

```
<!doctype>
<html>
<head>
<link rel="stylesheet" href="styles.css">
</head>
```



```
<body>
<h1> This is heading </h1>
<p> This is paragraph </p>
<p> Welcome </p>
</body>
</html>
```

- Example :-
css file :-

```
body
{
    background-color : #520066;
    text-align : center;
}

h2
{
    font-style : italic;
    size : 30px;
    color : #f08080;
}

p
{
    font-size : 20px;
}

.blue
{
    color : blue;
}

.red
{
    color : red;
}
```

```
.green
{
    color: green;
}
```

HTML file :-

```
<!doctype>
<html>
<head>
<link rel="stylesheet" type="text/css"
      href="hello1.css">
</head>
<body>
<h2> Learning HTML with CSS </h2>
<p class="blue"> Hello World </p>
<p class="green"> welcome </p>
<p class="red"> welcome to India </p>
</body>
</html>
```

- CSS style :-

```
<!doctype>
<html>
<head>
<style>
h1
{
    color: #ff0040;
    font-family: Times new roman;
    font-size: 300%;
}
```

```

p
{
    color : #006600;
    font-family : Times new roman;
    font-size : 160 %;
    border : 5px solid #520066;
    padding : 30px;
    margin : 50px;
}

```

```

</style>
</head>
<body style = "background-color:gray;" >
<h1> This is heading </h1>
<p> This is paragraph </p>
</body>
</html>

```

- Example :-

```

<!doctype>
<html>
<head>
<style>
h1
{
    color : crimson;
    font-family : times new roman;
    font-size : 300 %;
}
p
{
    color : darkslateBlue;
}

```

```
font-family : times new roman;
font-size : 160%;
border : 5px solid DarkMagenta;
padding : 30px;
margin : 50px;
```

```
</style>
</head>
<body style = "background-color : gray ;">
<h1> This is heading </h1>
<p> This is paragraph </p>
</body>
</html>
```

- Link to external CSS :-

External style sheets can be referenced with a full URL or with a path relative to the current web page.

Example :-

```
<!doctype>
<html>
<head>
<link rel = "stylesheet" href = "https://www.w3schools.com/html/style.css">
</head>
<body>
<h1> Hello world </h1>
<p> Welcome </p>
</body>
</html>
```


Links

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HTML Links are Hyperlinks.

You can click on a link and jump to another document.

When you move the mouse over a link, the mouse arrow will turn into a little hand.

Syntax :-

```
<a href = "url" > link text </a>
```

The most important attribute of the `<a>` element is the href attribute, which indicates the link destination.

Example :-

```
<!doctype>
<html>
<body>
<h1> HTML Links </h1>
<p><a href = "https://www.w3schools.com/" >
    Visit W3schools.com </a> </p>
</body>
</html>
```

By default links will appear as follows in all browsers :-

An universal link is underlined and blue.

A visited link is underlined and purple.

An active link is underlined and red.

- target attribute :-

Target attribute can have one of the following values.

- self :-

default. opens the document in the same window / tab as it was clicked.

- blank :-

opens the document in new window or tab.

- parent :-

opens the document in a new window or tab.

- top :-

opens the document in the full body of window.

- Example :-

```
<!doctype>
```

```
<html>
```

```
<body>
```

```
<h2> The target Attribute </h2>
```

```
<a href = "https://www.w3schools.com/" target = "-blank">
```

```
Visit W3schools </a>
```

```
</body>
```

```
</html>
```

- Absolute URLs VS Relative URLs :-

Absolute URL contain full web address, in the href attribute.

local link [a link to a page within the same website] is specified with a relative URL [without the "https://www" part].

Example :-

```
<!doctype>
```

```
<html>
```

```
<body>
```

```
<h2> Absolute URLs </h2>
```

```
<p><a href = "https://www.w3.org/">W3C</a></p>
<p><a href = "https://www.google.com/">Google</a></p>
<h2>Relative URLs</h2>
<p><a href = "styles.css">css code</a></p>
</body>
</html>
```

- Use image as a link :-

```
<!doctype>
<html>
<body>
<h2>Image as a link</h2>
<a href = "https://www.google.com"><img src = img.jpg"
alt = "HTML" style = "width:42px; height:42px;" />
</a>

</body>
</html>
```

- Link to an email address :-

```
<!doctype>
<html>
<body>
<h2>Link to an Email Address</h2>
<p><a href = "mailto:someone@example.com">
send email</a></p>

</body>
</html>
```

- Button as a link :-

```
<!doctype>
<html>
```

```

<body>
<h2> Button as a link </h2>
<button onclick = "document.location = 'https://www.
google.com'" > Google </button>
</body>
</html>

```

- Link title :-

```

<!doctype>
<html> lang = "en-US" >
<body>
<h2> Link Titles </h2>
<a href = "https://www.w3schools.com/html/"
title = "Go to W3 schools HTML section" >
Visit our HTML tutorial </a>
</body>
</html>

```

- Link color :-

By default,

- An unvisited link is underlined and blue.
- A visited link is underlined and purple.
- An active link is underlined and red.

Example :-

```

<!doctype>
<html>
<head>
<style>
a: link
{

```

```

color: green;

```



```
background-color: transparent;
text-decoration: none;
}
```

```
a: visited
```

```
{
```

```
color: pink;
background-color: transparent;
text-decoration: none;
```

```
}
```

```
a: hover
```

```
{
```

```
color: red;
background-color: transparent;
text-decoration: underline;
```

```
}
```

```
a: active
```

```
{
```

```
color: yellow;
background-color: transparent;
text-decoration: underline;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2> Link colors </h2>
```

```
<a href = "https://www.w3schools.com/html-images-
asp" target = "-blank"></a>
```

```
</body>
```

```
</html>
```

Link as button :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
a : link, a : Visited
```

```
{
```

```
background-color : #f44336;
```

```
color : white;
```

```
padding : 15px 25px;
```

```
text-align : center;
```

```
text-decoration : none;
```

```
display : inline-block;
```

```
}
```

```
a : hover, a : active
```

```
{
```

```
background-color : red;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2> Link Button </h2>
```

```
<a href = "https://www.gmail.com" target = "-blank">
```

```
This is a link </a>
```

```
</body>
```

```
</html>
```

- Link bookmark :-

It can be useful if web page is very long.

To create a bookmark -

First create bookmark, then add link to it.

Link as button :-

```
<!doctype>
<html>
<head>
<style>
a: link, a: Visited
{
    background-color: #f44336;
    color: white;
    padding: 15px 25px;
    text-align: center;
    text-decoration: none;
    display: inline-block;
}
a: hover, a: active
{
    background-color: red;
}
</style>
</head>
<body>
<h2> Link Button </h2>
<a href = "https://www.gmail.com" target = "-blank">
This is a link </a>
</body>
</html>
```

- Link bookmark :-

It can be useful if web page is very long.

To create a bookmark -

First create bookmark, then add link to it.

When link is clicked the page will scroll down or up to the location with the bookmark.

Example:-

```
<!doctype>
<html>
<body>
<p><a href = "#c4" > Jump to chapter 4 </a></p>
<p><a href = "#c10" > Jump to chapter 10 </a></p>
<h2> chapter 1 </h2>
<h2> chapter 2 </h2>
<h3> chapter 3 </h3>
<h2 id = "c4" > chapter 4 </h2>
<h2> chapter 5 </h2>
-----
<h2 id = "c10" > chapter 10 </h2>
<h2> chapter 11 </h2>
-----
<h2> chapter 27 </h2>
</body>
</html>
```


Image

Image can improve the design and appearance of web page.

- ** tag :-**

It is used to embed an image in a web page.

Image tag does not have closing tag.

Image tag required two attribute :-

src :-

Specifies the path to the image.

alt :-

Specifies an alternate text for the image.

Syntax :-

```
<img src = "url" alt = "alternatetext" >
```

- **Example :-**

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<title> Image tag </title>
```

```
<style>
```

```
body
```

```
{
```

```
background-color: gray;
```

```
}
```

```
img
```

```
{
```

```
border: 5px solid black;
```

```
}
```

```
</style>
```

```
</head>
```

```

<body>
<h2> HTML Image Example </h2>
<img src = "xyz.jpg" height = "180" width = "300" alt =
    "Image">
<img src = "animal.jpg" height = "180" width = "300" alt =
    "Image" style = "float: right;">
<br>
<br>
<img src = "abc.jpg" height = "180" width = "300" alt =
    "Image" style = "margin-left: auto;
    margin-right: auto; display: block;">
<br>
<br>
<img src = "img.jpg" height = "180" width = "300" alt =
    "Image">
<img src = "pqr.jpg" height = "180" width = "300" alt =
    "Image" style = "float: right;">
</body>
</html>

```

- Image in another folder :-

```

<!doctype>
<html>
<body>
<h2> Image in another folder </h2>
<img src = "C:\Users\NISHA\Desktop\Java\html5.jpg"
    alt = "HTML ICON" style = "width: 128px; height: 128px;">
</body>
</html>

```

- Image on another server or website :-

```
<!doctype>
<html>
<body>
<h2> Image on Another server </h2>

</body>
</html>
```

- Animated image :-

```
<!doctype>
<body>
<h2> Animated Image </h2>

</body>
</html>
```

- Image as a link :-

```
<!doctype>
<html>
<body>
<h2> Image as a link </h2>
<a href="https://www.google.com">
 </a>
</body>
</html>
```

Background image :-

```
<html>
<body>
<h2> Background Image </h2>
<p> Background image for div element </p>
<div style="background-image: url('img.jpg');">
    ----- write paragraph here ----- </div>

<p> Background image for p element </p>
<p style="background-image: url('img.jpg');">
    ----- Write paragraph here ----- </p>
</body>
</html>
```

Example :-

```
<!doctype>
<html>
<head>
<style>
div
{
    background-image: url('img.jpg');
}
</style>
</head>
<h2> Background image </h2>
<div> ----- Write paragraph here ----- </div>
</body>
</html>
```


- Background-image on page :-

```

<!doctype>
<html>
<head>
<style>
body
{
    background-image: url('coffee.jpg');
}
</style>
</head>
<body>
<h2 style="color: white;"> Background Image </h2>
<p style="color: white;"> -----
    ----- Write paragraph here ----- </p>
</body>
</html>

```

- Background no repeat :-

```

<!doctype>
<html>
<head>
<style>
body
{
    background-image: url('img.jpg');
    background-repeat: no repeat;
}
</style>
</head>
<body>

```

```
<h2> Background no repeat </h2>
```

```
<p> ----- Write code here ----- </p>
```

```
</body>
```

```
</html>
```

- Background - Cover :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
body
```

```
{
```

```
background-image : url ('img.jpg');
```

```
background-repeat : no-repeat;
```

```
background-attachment : fixed;
```

```
background-size : cover;
```

```
}
```

```
</style>
```

```
</head>
```

```
</body>
```

```
<h2> Background cover </h2>
```

```
<p> ----- Write paragraph here ----- </p>
```

```
</body>
```

```
</html>
```

- Background stretch :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
body  
{
```

```
background-image: url('pqx.jpg');
```

```
background-repeat: no-repeat;
```

```
background-attachment: fixed;
```

```
background-size: 100% 100%;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2> Background stretch </h2>
```

```
<p> ----- Write paragraph here </p>
```

```
</body>
```

```
</html>
```

Tables

Page No.	
Date	

`<table>` tag defines an HTML table.

Each table row is defined with a `<tr>` tag. Each table header is defined with a `<th>` tag.

Each table data/cell is defined with a `<td>` tag.

By default,

text in the `<th>` elements are bold and centered, regular and left-aligned.

`<td>` elements are data containers of the table.

Example :-

```
<!doctype>
<html>
<body>
<h2> Basic HTML table </h2>
<table style = "width : 100%">
  <tr>
    <th> First Name </th>
    <th> Last Name </th>
    <th> Age </th>
  </tr>
  <tr>
    <td> ABC </td>
    <td> abc </td>
    <td> 50 </td>
  </tr>
  <tr>
    <td> PQR </td>
    <td> pqr </td>
    <td> 60 </td>
  </tr>
```



```

<tr>
  <td> XYZ </td>
  <td> xyz </td>
  <td> 80 </td>
</tr>
</table>
</body>
</html>

```

- Add a border :-

```

<!doctype>
<html>
<head>
<style>
table, th, td
{
  border : 1px solid black;
}
</style>
</head>
<body>
<table style = "width : 100%">
<tr>
  <th> First Name </th>
  <th> Last Name </th>
  <th> Age </th>
</tr>
<tr>
  <td> Disha </td>
  <td> computer </td>
  <td> 25 </td>
</tr>

```

</table>

</body>

</html>

- Collaps border :-

<!doctype>

<html>

<head>

<style>

table th, td

{

border : 1px solid black;

border-collapse : collapse;

}

</style>

</head>

<body>

<table style = "width : 100%">

<tr>

<th> First Name </th>

<th> last Name </th>

</tr>

<tr>

<td> Disha Computer </td>

<td> xyz </td>

</tr>

</table>

</body>

</html>

- Cell padding :-

< head >

< style >

table, th, td

{

border : 3px solid black;

border-collapse : collapse;

}

th, td

{

padding : 15px;

}

< /style >

< /head >

- left-align heading :-

< head >

< style >

table, th, td

{

border : 3px solid black;

border-collapse : collapse;

}

th, td

{

padding : 15px;

}

th

{

text-align : left;

}

< /style >

< /head >

- Border spacing :-

<head>

<style>

table, th, td

{

border : 1px solid black;

padding : 5px ;

}

table

{

border-spacing : 15px ;

}

</style>

</head>

- Cell that span many columns :-

<table style = "width : 100%">

<tr><th> Name </th>

<th colspan = "2"> Telephone </th>

</tr>

<tr>

<td> Bill Gates </td>

<td> 123456789 </td>

<td> 987654321 </td>

</tr>

</table>

- Cell that span many rows :-

<table style = "width : 100%">

<tr>

<th> Name </th>

<td> Bill Gates </td>

</tr>


```
<tr>
  <th rowspan = "2" > Telephone </th>
  <td> 123456789 </td>
</tr>
<tr>
  <td> 987654321 </td>
</tr>
</table>
```

- Add a caption :-

```
<body>
<table style = "width: 100%">
<caption> Table </caption>
<tr>
  <th> Data </th>
  <th> Data </th>
  <th> Data </th>
</tr>
<tr>
  <td> 1 </td>
  <td> 2 </td>
  <td> 3 </td>
</tr>
</table>
</body>
```

- Special style for one table :-

```
<!doctype>
<html>
<head>
<style>
```

table, th, td

{

border : 1px solid black;

border-collapse : collapse;

}

th, td

{

padding : 15px;

text-align : left;

}

#t01

{

width : 100%;

background-color : pink;

}

</style>

<thead>

<tbody>

<table style = "width : 100%">

<tr>

<th> Data 1 </th>

<th> Data 1 </th>

<td> Data 1 </td>

</tr>

<tr>

<td> 1 </td>

<td> 2 </td>

<td> 3 </td>

</tr>

</table>

</body>

```

<table id="tbl">
  <tr>
    <th> Data 2 </th>
    <th> Data 2 </th>
    <th> Data 2 </th>
  </tr>
  <tr>
    <td> 1 </td>
    <td> 2 </td>
    <td> 3 </td>
  </tr>
</table>
</body>
</html>

```

- Add more styles:-

```

<!doctype>
<html>
<head>
<style>
table
{
  width: 100%;
}
table, th, td
{
  border: 1px solid black;
  border-collapse: collapse;
}
th, td
{
  padding: 15px;
  text-align: left;
}

```

```
#+01 tr : nth-child (even)
{
```

```
background-color : #EC3A79;
}
```

```
#+01 tr : nth-child (odd)
{
```

```
background-color : #2E4053;
}
```

```
#+01 th
{
```

```
background-color : black;
color : white;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<table>
```

```
<tr>
```

```
<th> data1 </th>
```

```
<th> data1 </th>
```

```
</tr>
```

```
<tr>
```

```
<td> 1 </td>
```

```
<td> 2 </td>
```

```
</tr>
```

```
<tr>
```

```
<td> 3 </td>
```

```
<td> 4 </td>
```

```
</tr>
```

```
<tr>
```

```
<td> 5 </td>
```



```
<td> 6 </td>
</tr>
</table>
<br>
<table id="t01">
  <tr>
    <th> Data 2 </th>
    <th> Data 2 </th>
  </tr>
  <tr>
    <td> 11 </td>
    <td> 22 </td>
  </tr>
  <tr>
    <td> 33 </td>
    <td> 44 </td>
  </tr>
  <tr>
    <td> 55 </td>
    <td> 66 </td>
  </tr>
</table>
</body>
</html>
```

Style only for one table :-

```
<html>
<head>
<style>
table
```

}

font-family : arial , sans-serif;

border-collapse : collapse;

width : 100 %;

}

td, th

{

border : 1px solid #dddddd;

text-align : left;

padding : 8px;

}

tr : nth-child (even)

{

background-color : #dddddd;

}

</style>

</head>

</html>

List

Page No.

Date

It is used to specify list of information. All lists may contains one or more list elements. There are three types of list.

- ordered list or Numbered list [ol]
- Unordered list or Bulleted list [ul]
- Description list or definition list [dl].

- HTML ordered list or numbered list :-

The ordered list starts with tag and list items starts with tag.

Example :-

```
<!doctype>
```

```
<html>
```

```
<body>
```

```
<ol>
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> HTML </li>
```

```
<li> Java </li>
```

```
</ol>
```

```
</body>
```

```
</html>
```

- Unordered list or bulleted list :-

The unordered list starts with tag and list items starts with the tag.

Example :-

```
<!doctype>
```

```
<html>
```

```
<body>
<ul>
  <li> C </li>
  <li> C++ </li>
  <li> HTML </li>
  <li> Java </li>
</ul>
</body>
</html>
```

- Description list or definition list :-

This list is very appropriate when you want to present glossary, list of terms or other name value list.

<dl> tag defines the start of the list.
 <dt> tag defines the term.
 <dd> tag defines the term definition [description].

- Example :-

```
<!doctype>
<html>
<body>
<dl>
  <dt> C </dt>
  <dd> - It is structural or procedural language </dd>
  <dt> C++ </dt>
  <dd> - It is object oriented programming language </dd>
  <dt> HTML </dt>
  <dd> - It is Hyper Text Markup Language </dd>
</dl>
</body>
</html>
```


- Nested list :-

```
<!doctype>
```

```
<html>
```

```
<body>
```

```
<p> list of states with their capital </p>
```

```
<ol>
```

```
<li> Delhi
```

```
<ul>
```

```
<li> New delhi </li>
```

```
</ul>
```

```
</li>
```

```
<li> Haryana
```

```
<ul>
```

```
<li> chandigarh </li>
```

```
</ul>
```

```
</li>
```

```
<li> Gujrat
```

```
<ul>
```

```
<li> Gandhinagar </li>
```

```
</ul>
```

```
</li>
```

```
<li> Rajasthan
```

```
<ul>
```

```
<li> Jaipur </li>
```

```
</ul>
```

```
</li>
```

```
</ol>
```

```
</body>
```

```
</html>
```

- Ordered list or Numbered list :-

There are different types of numbered list.

i. Numeric number :-

This is the default type. In this type list items are numbered with numbers [1, 2, 3]. For displaying this list we used Type "1".

ii. Capital roman number :-

In this type list items are numbered with upper case roman numbers [I, II, III]. For displaying this list we used Type "I".

iii. Small roman number :-

In this list items are numbered with lower case roman numbers [i, ii, iii]. For displaying this we used Type "i".

iv. Capital alphabets :-

In this type list items are numbered with upper case letter [A, B, C]. For displaying this we used Type "A".

v. Small alphabets :-

In this type list items are numbered with lower case letter [a, b, c]. For displaying this we used Type "a".

- Example :-

```
<!doctype >
```

```
<html>
```

```
<body>
```

```
<h4> Ordered list with numbers </h4>
```

```
<ol type="I">
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> Java </li>
```

```
</ol>
```

```
<h4> ordered list with upper case letter </h4>
```

```
<ol type="A">
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> Java </li>
```

```
</ol>
```

```
<h4> order list with lower case letter </h4>
```

```
<ol type="a">
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> Java </li>
```

```
</ol>
```

```
<h4> ordered list with capital roman number </h4>
```

```
<ol type="I">
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> Java </li>
```

```
</ol>
```

```
<h4> ordered list with small roman number </h4>
```

```
<ol type="i">
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> Java </li>
```

```
</ol>
```

```
</body>
```

```
</html>
```

- List counting :-

By default ordered list start counting from 1. If you want to start counting from specified number, you can use start attribute.

Example :-

```
<!doctype>
<html>
<body>
<h2> start attribute </h2>
<ol start = "50">
  <li> C </li>
  <li> C++ </li>
  <li> Java </li>
</ol>
<ol type = "I" start = "50">
  <li> C </li>
  <li> C++ </li>
  <li> Java </li>
</ol>
</body>
</html>
```

- Nested list :- [list inside list]

```
<!doctype>
<html>
<body>
<h2> Nested list </h2>
<ol>
  <li> C </li>
  <li> Python
    <ol>
      <li> Core python </li>
```



```
<li> Advance Python </li>
```

```
</ol>
```

```
<li>
```

```
<li> Java </li>
```

```
</ol>
```

```
</body>
```

```
</html>
```

- Reversed list :-

```
<!doctype>
```

```
<html>
```

```
<body>
```

```
<h2> Reversed list </h2>
```

```
<ol reversed>
```

```
<li> HTML </li>
```

```
<li> Java </li>
```

```
<li> C++ </li>
```

```
<li> C </li>
```

```
</ol>
```

```
</body>
```

```
</html>
```

- Unordered list or bulleted list :-

```
<!doctype>
```

```
<html>
```

```
<body>
```

```
<h4> Unordered list with bullets </h4>
```

```
<ul style="list-style-type: disc;">
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> Python </li>
```

```
</ul>
```

<h4> Unordered list with circle bullets </h4>

```
<ul style="list-style-type: circle;">
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> Python </li>
```

```
</ul>
```

<h4> Unordered list with square bullets </h4>

```
<ul style="list-style-type: square;">
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> Python </li>
```

```
</ul>
```

<h4> Unordered list without bullets </h4>

```
<ul style="list-style-type: none;">
```

```
<li> C </li>
```

```
<li> C++ </li>
```

```
<li> Python </li>
```

```
</ul>
```

```
</body>
```

```
</html>
```

- Nested list :-

```
<!doctype>
```

```
<html>
```

```
<body>
```

```
<h2> Nested list </h2>
```

```
<ul>
```

```
<li> C </li>
```

```
<li> Python </li>
```

```
<ul>
```

```
<li> core python </li>
```

```
<li> Advance python </li>
```

```
</ul>
```

```

</li>
<li> C++ </li>
</ul>
</body>
</html>

```

- Horizontal list with CSS :-

List can be styled in many different ways with CSS. One popular way is to style a list horizontally, to create a navigation menu.

Example :-

```

<!doctype>
<html>
<head>
<style>
ul
{
    list-style-type : none;
    margin : 0;
    padding : 0;
    overflow : hidden;
    background-color : #333333;
}
li
{
    float : left;
}
li a
{
    display : block;
    color : white;

```

```

text-align: center;
padding: 16px;
text-decoration: none;
}
li a: hover
{
background-color: #111111;
}

```

```

</style>

```

```

<head>

```

```

<body>

```

```

<h2> Navigation menu </h2>

```

```

<ul>

```

```

<li><a href = "#home"> Home </a></li>

```

```

<li><a href = "#news"> News </a></li>

```

```

<li><a href = "#contacts"> Contact </a></li>

```

```

<li><a href = "#about"> About </a></li>

```

```

</ul>

```

```

</body>

```

```

</html>

```


class

class attribute is used to specify the class for HTML elements.

Multiple HTML elements can share same class.

- Using class attribute :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
body
```

```
{
```

```
background-color: gray;
```

```
}
```

```
hello
```

```
{
```

```
background-color: #FF0066;
```

```
color: white;
```

```
border: 2px solid black;
```

```
margin: 20px;
```

```
padding: 20px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class = "hello">
```

```
<h2> C </h2>
```

```
<p> C is the structural programming language </p>
```

```
</div>
```

```
<div class = "hello">
```

```
<h2> C++ </h2>
```

```
<p> C++ is object oriented Programming language </p>
```

```
</div>
```

```
<div class = "hello" >
<h2> HTML </h2>
<p> HTML is the Hyper Text Markup Language. </p>
</div>
</body>
</html>
```

- Example :-

```
<!doctype>
<html>
<head>
<style>
    .note
    {
        font-size : 120%;
        color : red;
    }
</style>
</head>
<body>
<h1> my <span class = "note" > Important </span>
        Heading. </h1>
<p> This is some <span class = "note" > important </span>
        text. </p>
<h1> my <mark class = "note" > Important </mark>
        Heading </h1>
<p> This is some <mark class = "note" > important </mark>
        text </p>
</body>
</html>
```

- Example :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
body
```

```
{
```

```
background-color: gray;
```

```
}
```

```
·hello
```

```
{
```

```
background-color: #FF0066;
```

```
color: white;
```

```
padding: 1px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2> The class attribute </h2>
```

```
<h2 class="hello"> C </h2>
```

```
<p> C is a structural language </p>
```

```
<h2 class="hello"> C++ </h2>
```

```
<p> C++ is an object oriented programming language </p>
```

```
<h2 class="hello"> HTML </h2>
```

```
<p> HTML is Hyper Text Markup language </p>
```

```
</body>
```

```
</html>
```

- Example :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```

<style>
body
{
    background-color: gray;
}
.hello
{
    background-color: #FF0066;
    color: white;
    padding: 10px;
}

```

```

</style>
<head>
<body>
<h2> The class attribute </h2>
<h2 class="hello"> C </h2>
<p> C is a structural language </p>
<h2 class="hello"> C++ </h2>
<p> C++ is object oriented programming language </p>
<h2 class="hello"> HTML </h2>
<p> HTML is Hyper Text Markup language </p>
</body>
</html>

```

- Multiple classes :-

HTML elements can belong to more than one class.

To define multiple classes, separate the class names with space, e.g. <div class="city main"> The element will be styled according to all the classes specified.

Example :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
body
```

```
{
```

```
background-color: gray;
```

```
}
```

```
.city
```

```
{
```

```
background-color: #FF0066;
```

```
color: white;
```

```
padding: 10px;
```

```
}
```

```
.main
```

```
{
```

```
text-align: center;
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2> Multiple classes </h2>
```

```
<h2 class = "city main"> London </h2>
```

```
<h2 class = "city"> Paris </h2>
```

```
<h2 class = "city"> New York </h2>
```

```
</body>
```

```
</html>
```

- Different element can share same class :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```

<style>
body
{
    background-color: gray;
}
.hello
{
    background-color: #FF0066;
    color: white;
    padding: 10px;
}

```

```

</style>
</head>
<body>
<h2> Different element can share same class </h2>
<h2 class = "hello"> HTML </h2>
<p class = "hello"> HTML stands for Hyper Text
    Markup language </p>
</body>
</html>

```

- Use the class attribute in javascript :-
class name can also be used by javascript to perform certain task for specific elements.
JavaScript can access elements with a getElementByClassName() method.

- Example :-

```

<!doctype>
<html>
<body>
<h2> class attribute used in Javascript </h2>

```

```
<button onclick = "myfunction()" > Hide element </button>
<h2 class = "city" > London </h2>
<p> London is capital of England. </p>
<h2 class = "city" > Paris </h2>
<p> Paris is capital of France </p>
<h2 class = "city" > Tokyo </h2>
<p> Tokyo is capital of Japan </p>
<script>
function myfunction ()
{
    var x = document.getElementsByClassName ("city");
    for (var i = 0; i < x.length; i++)
    {
        x[i].style.display = "none";
    }
}
</script>
</body>
</html>
```

id attribute.

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It is used to specify unique id for HTML element.

You cannot have more than one element with the same id in HTML document.

id attribute used to point a specific style declaration in a style sheet.

Syntax :-

Write a hash character [#], followed by an id name.

Then defines the CSS properties within curly braces {}.

Example :-

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
# myHeader
```

```
{
```

```
background-color : lightblue;
```

```
color : black;
```

```
padding : 40px;
```

```
text-align : center;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2 id attribute </h2>
```

```
<h1 id = "myHeader" > My Header. </h1>
```

```
</body>
```

```
</html>
```

Difference between class and id :-

A class name can be used by multiple HTML elements, while

an id name must only be used by one HTML element within the page.

- Example :-

```
<!doctype>
<html>
<head>
<style>
# myHeader
{
    background-color : green;
    color : black;
    padding : 4px;
    text-align : center;
    border : 4px solid black;
}
.city
{
    background-color : #FF0066;
    color : white;
    padding : 10px;
    border : 3px solid black;
}
body
{
    background-color : gray;
}
</style>
</head>
<body>
<h2> Difference between class and Id </h2>
```

```
<h1 id = "myHeader" > my cities </h1>
<h2 class = "city" > London </h2>
<p> London is the capital of England </p>
<h2 class = "city" > Paris </h2>
<p> Paris is the capital of France </p>
<h2 class = "city" > Tokyo </h2>
<p> Tokyo is the capital of Japan </p>
</body>
</html>
```

- Multiple Id's :-

```
<!doctype>
<html>
<head>
<title> Id attribute in css </title>
<style>
# cars
{
padding : 40px;
background-color : # FF0066;
color : black;
text-align : center;
}
# Bikes
{
padding : 50px;
background-color : # 000000;
text-align : center;
color : white;
}
</style>
</head>
```

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```
<body style="background-color:gray;">
<h1 id="cars">cars </h1>
<h1 id="Bikes"> Bikes </h1>
</body>
</html>
```

- Id attribute in javascript :-

id attribute can also be used by Javascript to perform some specific task.

Javascript can access an element with specific id with the getElementById() method.

Example :-

```
<!doctype>
<html>
<body>
<h2> Id attribute using Javascript </h2>
<h1 id="myHeader"> Hello World </h1>
<button onclick="displayResult()"> change text </button>
<script>
function displayResult ()
{
    document.getElementById("myHeader").innerHTML
    = "Have a nice day";
}
</script>
</body>
</html>
```

Form

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It is a section of document which contains controls such as text fields, password fields, checkboxes, radio buttons, submit button, menus etc.

Why use HTML form :-

HTML forms are required if you want to collect some data from the site visitor.

Syntax of form :-

```
<form action = "server url" method = "get/post">  
  // input controls e.g. textfield, text area, radiobutton,  
  button  
</form>
```

- input element :-

```
<body>
```

```
<form>
```

```
  Enter your name <br>
```

```
  <input type = "text" name = "username">
```

```
</form>
```

```
</body>
```

- text field element :-

```
<form>
```

```
  First name : <input type = "text" name = "firstname">
```

```
  <br> <br>
```

```
  Last name : <input type = "text" name = "lastname">
```

```
  <br>
```

```
</form>
```


- textarea element :-

```
<!doctype>
<html>
<head>
  <title> Form in HTML </title>
</head>
<body>
  <form>
    Enter your address : <br>
    <textarea rows = "2" cols = "20"></textarea>
  </form>
</body>
</html>
```

- label tag :-

```
<form>
  <label for = "firstname"> First Name </label> <br>
  <input type = "text" id = "firstname" name =
    "firstname"> <br>
  <label for = "lastname"> Last Name </label> <br>
  <input type = "text" id = "lastname" name =
    "lastname"> <br>
</form>
```

- Password field :-

```
<form>
  <label for = "password"> Password : </label>
  <input type = "password" id = "password" name =
    "password"> <br>
</form>
```

- email field :-

```
<form>
  <label for = "email" > Email : </label>
  <input type = "email" id = "email" name = "email">
</form>
```

- Radio button control :-

```
<form>
  <label for = "exam" > Exam : </label>
  <input type = "radio" id = "exam" name = "exam"
    value = "male" > C
  <input type = "radio" id = "exam" name = "exam"
    value = "female" > C++
</form>
```

- checkbox control :-

```
<form>
Hobby : <br>
  <input type = "checkbox" id = "cricket" name = "cricket"
    value = "cricket" >
  <label for = "cricket" > cricket </label> <br>
  <input type = "checkbox" id = "football" name = "football"
    value = "football" >
  <label for = "football" > Football </label> <br>
  <input type = "checkbox" id = "hockey" name = "hockey"
    value = "hockey" >
  <label for = "hockey" > Hockey </label>
</form>
```

- submit button :-

```
<input type = "submit" value = "submit" >
```

- <fieldset> element :-

```
<form>
```

```
<fieldset>
```

```
<legend> User Information </legend>
```

```
<label for = "name"> Enter name </label> <br>
```

```
<input type = "text" id = "name" name = "name">
```

```
<label for = "pass"> Enter password </label> <br>
```

```
<input type = "password" id = "pass" name = "pass">
```

```
<br> <br>
```

```
</fieldset>
```

```
</form>
```

- Write a program to create registration form.

```
<!doctype>
```

```
<html>
```

```
<head>
```

```
<title> Form in HTML </title>
```

```
</head>
```

```
<body>
```

```
<h2> Registration form </h2>
```

```
<form>
```

```
<fieldset style = "border : 2px solid black ;">
```

```
<legend> User personal information </legend>
```

```
<label> Enter your full name </label> <br>
```

```
<input type = "text" name = "name"> <br>
```

```
<label> Enter your e-mail </label> <br>
```

```
<input type = "email" name = "email"> <br>
```

```
<label> Enter your password </label> <br>
```

```
<input type = "password" name = "pass"> <br>
```

```
<label> Confirm your password </label> <br>
```

```
<input type = "password" name = "pass"> <br> <br>
```

```
<label> Enter your exam type </label> <br>
<input type = "radio" id = "exam" name = "exam" value =
    "male" > Online <br>
<input type = "radio" id = "exam" name = "exam" value =
    "female" > Offline <br> <br>
Enter your address : <br>
<textarea> </textarea> <br> <br>
<input type = "submit" value = "sign-up" >
</fieldset>
</form>
</body>
</html>
```

- Example :-

```
<form action = "#" autocomplete = "off">
<table>
<tr>
<td class = "tdLabel"> <label for = "register-name"
    class = "label"> Enter name : </label> </td>
<td> <input type = "text" name = "name" value = ""
    id = "register-name" style = "width : 160px"> </td>
</tr>
<tr>
<td class = "tdLabel"> <label for = "register-password"
    class = "label"> Enter password : </label> </td>
<td> <input type = "password" name = "password"
    id = "register-password" style = "width : 160px"> </td>
</tr>
<tr>
<td class = "tdLabel"> <label for = "register-email"
    class = "label"> Enter Email : </label> </td>
```



```

<td><input type = "email" name = "email" value = " "
id = "register-email" style = "width: 160px" ></td>
</tr>
<tr>
<td class = "tdlabel"><label for = "register-gender"
class = "label">Enter gender : </label></td>
<td>
<input type = "radio" name = "gender" id = "male"
value = "male" >
<label for = "male"> male </label>
<input type = "radio" name = "gender" id = "female"
value = "female" >
<label for = "female"> female </label>
</td>
</tr>
<tr>
<td class = "tdlabel"><label for = "register-country"
class = "label">select country : </label></td>
<td>
<select name = "country" id = "register-country"
style = "width: 160px" >
<option value = "India"> India </option>
<option value = "America"> America </option>
<option value = "Japan"> Japan </option>
<option value = "london"> london </option>
<option value = "other"> other </option>
</select>
</td>
</tr>
<tr>
<td colspan = "2">

```

```
<div align="right"><input type="submit" id="register"
value="register">
```

```
</div>
```

```
</td>
```

```
</tr>
```

```
</table>
```

```
</form>
```

- reset element :-

```
<form>
```

```
<label> user id : </label>
```

```
<input type="text" name="user-id" value="">
```

```
<label> Password : </label>
```

```
<input type="password" name="pass" value="">
```

```
<br><br>
```

```
<input type="submit" value="login">
```

```
<input type="reset" value="Reset">
```

```
</form>
```

- <input type="file"> :-

```
<form>
```

```
<label> select file to upload </label>
```

```
<input type="file" name="newfile">
```

```
<input type="submit" value="submit">
```

```
</form>
```

- <input type="image">

```
<form>
```

```
<label> user Id : </label><br>
```

```
<input type="text" name="name"><br><br>
```

```
<input type="image" alt="submit" src="login.jpg"
```

```
width="100px">
```

```
</form>
```

- `<input type = "color" >`

`<form>`

`<input type = "color" name = "upclick" value = "#052a2a" > upclick <br>`

`<input type = "color" name = "downclick" value = "#f5f5dc" > Downclick`

`</form>`

- `<input type = "date" >`

`<form>`

`<input type = "date" name = "startdate" > Start Date: <br>`

`<input type = "date" name = "enddate" > End date: <br>`

`<input type = "submit" >`

`</form>`

- `<input type = "datetime-local" > :-`

`<form>`

`<label > select date and time </label>`

`<input type = "datetime-local" name = "meetingdate" >`

`<input type = "submit" >`

`</form>`

- `<input type = "email" >`

`<form>`

`<label > <b> Enter your Email address </b> </label>`

`<input type = "email" name = "email" required >`

`<input type = "submit" > <br> <br>`

`<label > <b> Enter multiple Email address </b> </label>`

`<input type = "email" name = "email" multiple >`

`<input type = "submit" >`

`</form>`

- `<input type = "month" > :-`

`<form >`

`<input type = "month" name = "newmonth" >`

`<input type = "submit" >`

`</form >`

- `<input type = "number" > :-`

`<form >`

`<label > Enter your age : </label >`

`<input type = "number" name = "age" min = "20" max = "50" >`

`<input type = "submit" >`

`</form >`

- `<input type = "url" > :-`

`<form >`

`<label > Enter your website URL : </label >`

`<input type = "url" name = "website" placeholder = "https:// example . com" > <br >`

`<input type = "submit" value = "send data" >`

`</form >`

- `<input type = "week" > :-`

`<form >`

`<label > select the week in the year </label > <br > <br >`

`<input type = "week" name = "weeks" >`

`<input type = "submit" value = "send data" >`

`</form >`

- `<input type = "search" > :-`

`<form >`

`<label > search here : </label >`


```
<input type = "search" name = "q">
<input type = "submit" value = "search">
</form>
```

- <input type = "tel" > :-

```
<form>
<label> Enter your telephone number (in format of
aaa-aaa-aaaa) : </label>
<input type = "tel" name = "telephone" pattern =
"[0-9]{3}-[0-9]{3}-[0-9]{4}" required >
<input type = "submit" > <br> <br>
</form>
```

- disable attribute :-

```
<form>
<label> Enter user name : </label> <br>
<input type = "text" name = "uname" value = "USER"
disabled > <br> <br>
<label> Enter your password : </label> <br>
<input type = "password" name = "pass" > <br> <br>
<input type = "submit" value = "login">
</form>
```

- size attribute :-

```
<label> Enter your name </label> <br>
<input type = "text" name = "uname" size = "40"
required > <br> <br>
```

Block and Inline element

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```
- <article>:-  
<head>  
<style>  
  .all-browsers  
  {  
    margin: 0;  
    padding: 5px;  
    background-color: gray;  
  }  
  .all-browser > h1, .browser  
  {  
    margin: 10px;  
    padding: 5px;  
  }  
  .browser  
  {  
    background-color: white;  
  }  
  .browser > h2, p  
  {  
    margin: 4px;  
    font-size: 90%;  
  }  
</style>  
</head>  
<h1> Article element - styled with CSS </h1>  
<article class = "all-browsers">  
<h1> Languages </h1>  
<article class = "browser">  
<h2> C </h2>  
<p> C is a structural / procedural language </p>  
</article>
```

```
<article class = "browser">
<h2> HTML </h2>
<p> Hyper Text Markup Language </p>
</article>
</article>
```

- <aside> :-

```
<head>
<style>
aside
{
    width : 30 %;
    padding - left : 15px;
    margin - left : 15px;
    float : right;
    font - style : italic;
    background - color : lightgray;
}
```

```
</style>
</head>
<body>
<p> ----- </p>
<aside> ----- </aside>
</body>
```

- <canvas> :-

```
<canvas id = "mycanvas"> </canvas>
<script>
var c = document.getElementById("mycanvas");
var ctx = c.getContext("2d");
ctx.fillStyle = "FF0066";
ctx.fillRect(0,0,80,100);
</script>
```

Example:-

```
<canvas id = "mycanvas" ></canvas>
<script>
var c = document.getElementById("mycanvas");
var ctx = c.getContext("2d");
ctx.fillStyle = "#FF0066";
ctx.fillRect(20, 20, 75, 50);
// For transparency
ctx.globalAlpha = 0.6;
ctx.fillStyle = "#4E0858";
ctx.fillRect(50, 50, 75, 50);
ctx.fillStyle = "#134361";
ctx.fillRect(80, 80, 75, 50);
</script>
```

- <div> :-

```
<!doctype>
<html>
<style>
.myDiv
{
border : 5px outset black;
background-color : #FF0066;
text-align : center;
}
</style>
<head>
<body>
<h1> Div element </h1>
<div class = "myDiv">
<h2> This is heading </h2> </div>
</body>
</html>
```


- <figure> and <figcaption> :-

```
<!doctype>
<html>
<body>
<h1> The figure and figcaption element </h1>
<figure>
  
  <figcaption> Fig1 - Image of flower </figcaption>
</figure>
</body>
</html>
```

- <hr> :-

```
<!doctype>
<html>
<body>
<h4> HTML </h4>
<hr>
<h4> CSS </h4>
<hr style="height: 2px; border-width: 0; color: gray;
background-color: gray;" />
</body>
</html>
```

- <nav> :-

```
<!doctype>
<html>
<body>
<h1> Navigation element </h1>
<nav>
<a href="https://www.google.com"> Google </a> |
```

```
<a href = "https://www.facebook.com"> Facebook </a> |
<a href = "https://www.twitter.com"> Twitter </a> |
<a href = "https://www.instagram.com"> Instagram </a>
</nav>
</body>
</html>
```

```
- <noscript>
<!doctype>
<html>
<body>
<script>
document.write("Hello World")
</script>
<noscript> Welcome to India </noscript>
</body>
</html>
```

```
- <pre>
<!doctype>
<html>
<body>
<h2> standard pre </h2>
pre> ----- write text ----- </pre>
<h2> Fixed width pre </h2>
<div style = "width : 200px ; overflow : auto">
<pre> ----- write text ----- </pre>
</div>
</body>
</html>
```

<dfn> :- Definition element

`<!doctype>`

`<html>`

`<body>`

`<h1> The definition element <dfn> <h1>`

`<p> <dfn> HTML <dfn> is the standard markup language for creating web pages </p>`

`</body>`

`</html>`

<object> :-

`<!doctype>`

`<html>`

`<body>`

`<object data = "img1.jpg" width = "300" height = "200">`
`</object>`

`<object data = "movie.mp4" width = "400" height = "300">`
`</object>`

`</body>`

`</html>`

`<!doctype>`

`<html>`

`<body>`

`<h1> The span element <h1>`

`<p> Welcome to <span style = "color: blue; font-weight: bold"> Disha Computer </span> Institute`

`<span style = "color: #FF0066; font-weight: bold">`
`Ahmednagar, </span> 414003 </p>`

`</body>`

`</html>`